EE6094
CAD for VLSI Design

# Hint

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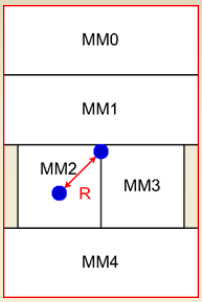
## Problem description

- Place modules without overlap while minimizing the bounding box area and Integral Nonlinearity (INL).
- Input
  - A set of rectangular modules(transistors)
- Output
  - A legal floorplan result
- Objective
  - Minimize the bounding box area while considering Integral Nonlinearity

# Output format

```
72.4548
4.99 14.52
7.53
MM0 0.0 12.4 (4.99 2.12 4 1)
MM1 0.0 10.28 (4.99 2.12 4 1)
MM2 0.595 5.54 (1.9 4.74 1 1)
MM3 2.495 5.54 (1.9 4.74 1 1)
MM4 0.0 0.0 (4.99 5.54 4 1)
```




<chip area>
<chip width> <chip height>
<INL>
<device_name><x><y>(<width><height><col_multiple>
<row_multiple>)



# Question

- Q1 : Case2計算INL部分無法通過checker測試原因為小數點後的值有些許差異
- Q2 : INL的輸出要求四捨五入到小數點第二位，但checker 報錯時算到小數點第四位

# Checker

◆Example

➢ If your result's INL does not match your floorplan result, you will get the "**INL mismatch**" message.

```
2061.9880
37.22 55.40
200.21
MM0 0.00 0.00 (9.11 5.54 8 1)
MM1 9.11 0.00 (9.11 5.54 8 1)
MM2 18.22 0.00 (1.90 4.74 1 1)
MM3 20.12 0.00 (1.90 4.74 1 1)
MM4 22.02 0.00 (1.90 18.96 1 4)
MM5 23.92 0.00 (1.90 18.96 1 4)
MM6 25.82 0.00 (1.90 8.48 1 2)
MM7 27.72 0.00 (1.90 8.48 1 2)
MM8 29.62 0.00 (1.90 8.48 1 2)
MM9 31.52 0.00 (1.90 8.48 1 2)
MM10 33.42 0.00 (1.90 55.40 1 10)
MM11 35.32 0.00 (1.90 11.08 1 2)
```

case1.out

```
Error: INL mismatch. Calculated: 226.211, Reported: 200.21
```

result



# Hint

◆Checker檢測可容許誤差小於1E-02

◆計算INL時，最後輸出再做四捨五入到小數點下二位即可

◆計算INL時，需對block名稱由小到大做排序